

Francisco Costela, PhD

Machine Learning Engineer

Sony Electronics

 francisco-costela

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Professional summary

I have a strong passion to work with data, create ETL pipelines, build and deploy AI models, and run quantitative analysis. 10+ years professional programming. Wide experience with AI - Object Recognition Models, Large Language Models, AI Deployment on Sensors, Data Modeling - as well as Experimental Design Testing, Technical Literacy, and versatile Project Management adhering to Agile Methodologies.

Relevant skills

AI/ML	Deep Learning, TensorFlow, Keras, Detectors (YOLO), NLP, Link to my Portfolio 
LLM	Built RAG assisted LLM chatbot powered by Azure OpenAI, Prompt Engineering, HyDE
Data Science	Python, Numpy, Pandas, SQL, PostgreSQL, PgAdmin, Rclone, PM2, HTK, Agile/Scrum
AI platforms	Azure OpenAI, Custom Vision, Edge Impulse, Ultralytics, Neurala Brainbuilder, Netspresso
Coding	Fluent in Python, Conda, Matlab. Knowledge of QT, C/C++, Java  Github link
Research	21 peer reviewed articles  Google Scholar link - 31 verified reviews  Publons link
Communication	22 contributions to international meetings, including ICMLT, ARVO, SFN, VSS

Employment

- 2022-present **Applied Machine Learning Engineer**, at *Sony Electronics*, San Diego, CA.
- Created, deployed, and tested AI models on IoT/Edge AI image Sony IMX500 sensors using Sony AITRIOS, cloud processing and API tools, Python, and Azure services
 - Wrote technical tutorials integrating Sony IMX500 with AI models from multiple AI platforms
 - Applications: RAG assisted LLM chatbot to provide real-time inventory data and insight generation, Retail on-shelf availability, Price tag verification, Worker station and customer line tracking, Gaze and demographic detection for dynamic advertising applications
- 2020-2022 **Sr Data Scientist**, at *Sony Electronics*, San Diego, CA.
- Built Database architecture to manage/store TBs of data in Visual Speech Recognition project
 - Built Audio and Video Processing ETL pipeline performing forced alignment, speech recognition, and face detection using Python, OpenCV, and HTK
 - Built, deployed, and tested Object Detection Models using Custom Vision and TensorFlow
- 2019-2020 **ML Innovation Researcher**, at *Harvard Medical School*, Boston, MA.
- Predicted hazard detection using eye tracking with feed-forward neural networks
 - Conducted NLP analyses from auditory descriptions using Amazon Mechanical Turk
 - Predicted gazes in gaze-contingent paradigms using Taylor series and RNNs
- 2015-2020 **Data Scientist**, as *Sr Fellow at Schepens Eye Research Institute*, Boston, MA.
- Built a freely open-access eye-movement database watching movies. [Link](#) 
 - Applied gaze tracking and video processing to identify the preferred retinal locus used by people with low vision while watching video
 - Applied unsupervised learning methods (clustering and archetypal analysis) to classify visual functional fields in a population with retinitis pigmentosa
 - Created rehabilitation aids in low vision (i.e. macular degeneration and hemianopia)

- 2011-2014 **Computer Vision Researcher**, at *Barrow Neurological Institute*, Phoenix, AZ.
 - Applied electrophysiology with macaques to map eye movements in the visual cortex
 - Investigated the roles of fixational eye movements in perception, cognition, and oculomotor control developing psychophysics experiments
 - Developed software to import, process, and plot eye tracking data
- 2009-2010 **Medical Imaging Researcher**, at *Pablo de Olavide University*, Seville, Spain.
 - Analysed fiber tract and DTI (Diffusion tensor imaging) from patients with dementia
 - Processed MRI data using Freesurfer pipelines and FSL segmentation tools
- 2009 **Medical Imaging Researcher**, as *Trainee at FMRI Oxford University*, Oxford, UK.
 - Detected the stria of Gennari (area in the visual cortex) in brain MRI scans

Education

- 2011-2014 **PhD in Neuroscience**, *Arizona State university*, Tempe, AZ, USA.
- 2019-2010 **MSc in Neuroscience**, *Pablo de Olavide University*, Seville, Spain.
- 2008-2009 **MSc in Biomedical Engineering**, *Oxford University*, Oxford, UK.
- 2005-2006 **BSc Computer Engineering - thesis**, *Heriot-Watt University*, Edinburgh, UK.
- 2000-2005 **BSc Computer Engineering**, *University of Granada*, Granada, Spain.

Patents

- SYP354081US02 **Visual Speech Recognition Based On Lip Movements Using Generative Artificial Intelligence (AI) Model.**
- SYP354081US01 **Visual Speech Recognition for Understanding Unstructured Sentences.**

Relevant courses

- 2020-2024 **ML/DL/LLM courses**, *Convolutional Neural Networks in TensorFlow, Improving Deep Neural Networks, Structuring Machine Learning Projects, Neural Networks and Deep Learning, ChatGPT Prompt Engineering for Developers, Coursera DeepLearning.AI.*
- 2017 **Effectively Communicating Research**, *Harvard Catalyst.*
- 2015-2016 **Certificate in Applied Biostatistics**, *Harvard Catalyst.*
- 2015 **Fundamentals of Clinical and Translational Research**, *Harvard Catalyst.*
- 2013 **Machine Learning**, *Coursera Stanford University.*

Other work experience

- 2017-2020 **"Vision: Eye Movements"**, *Lecturer, Harvard University/SERI*, 2 semesters.
- 2015-2022 **"IMFAHE Fellowship"**, *Advisor, Spain-USA*, 10 grad students.
- 2015-2022 **"International Mentor Program"**, *Mentor, Spain-USA*, 4 grad students.
- 2013-2014 **"BIO 201 Anatomy and Physiology"**, *TA, SLS, ASU*, 3 semesters.
- 2007–2008 **Senior Programmer**, *Ebrisa, Chicago*, MyEclipse, J2EE, MySQL, CSS, Taglibs, XML.
- 2006–2007 **Junior Programmer**, *Telefonica I+D, Madrid, Spain*, JAVA Technology, J2EE, JABX, SOAP, XML, Javascript, Beanshells, PJTools, Oracle Database.
- 2005–2006 **Junior Programmer**, *Ocean Lab, Edinburgh, UK*, Design of an autonomous underwater vehicle. Sonar management, C++, OpenCV.
- 2005 **Trainee**, *MADOC, Granada, Spain*, Maintenance and development of applications involving database access to Oracle and Access from Delphi and Jbuilder.